MARKING SCHEME

Model Paper — 01

Mathematics II

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Notes (for teachers):

- Candidates answer any five questions from Part A (Questions 1 – 6) and any five from Part B (Questions 7 – 12). Each question carries 10 marks. Total 100.
- The marks for each step are shown in the right-hand column and add up to the marks printed for each part of the question paper.
- Give full marks for correct alternative methods. Accept reasonable ranges for values read from graphs, tables and drawings.

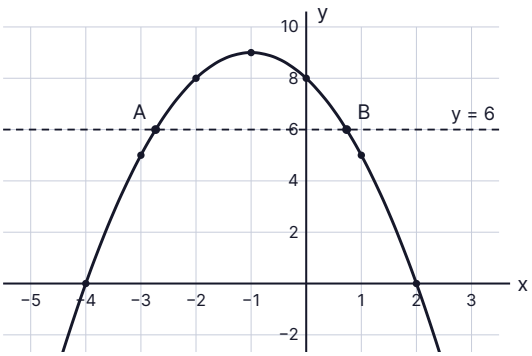
Part A — Answers

Question 1

(i)	
Interest = $48\,000 \times \frac{10}{100} \times 1$ = Rs 4 800	**1**
Total paid = 48 000 + 4 800 = Rs 52 800	**1**
(ii)	
Instalment = 51 120 ÷ 12 = Rs 4 260	**1**
(iii)	
Each month Rs 4 000 of the price is repaid, so the balances are 48 000, 44 000, 40 000, …, 4 000 (12 terms)	**1**
$= 4\,000 \times (12 + 11 + \ldots + 1) = 4\,000 \times \frac{12 \times 13}{2}$	**1**
$= 4\,000 \times 78$ = Rs 312 000 (or the sum of an AP: $\frac{12}{2}(48\,000 + 4\,000) = 312\,000$)	**1**
(iv)	
$3\,120 = 312\,000 \times \frac{r}{100} \times \frac{1}{12}$	**1**
$r = \frac{3\,120 \times 100 \times 12}{312\,000} = 12$	**2**
Annual rate under Method 2 = 12% > 10%, so it is higher than under Method 1.	**1**

Total marks: 10

Question 2

(i)	
By symmetry about x = −1, the value at x = 0 equals the value at x = −2: y = 8	**1**
(ii)	
 A correct graph (shown reduced in size)	
Suitable scale on both axes, axes labelled	**1**
All 9 points plotted correctly (8 or 9: 1 mark; fewer than 8: 0)	**1**
Smooth curve, maximum point (−1, 9)	**1**
(iii)	
(a) x = −4 and x = 2	**1**
(b) y = −(x + 4)(x − 2)	**1**
(iv)	
(a) The line y = 6 drawn; A and B at x ≈ −2.7 and x ≈ 0.7 (accept −2.8 to −2.6 and 0.6 to 0.8)	**1**
y > 6 for −2.7 < x < 0.7	**1**
(b) AB = 0.7 − (−2.7) = 3.4 to 3.5 (exact: x = −1 ± √3, AB = 2√3 = 3.46 ≈ 3.5; accept 3.4 – 3.6)	**2**

Total marks: 10

Question 3

(i)	
In $\triangle ABC$: $BC = AC \sin 20°40' = 45 \times 0.3529$	**1**
$BC = 15.88$ m (from $\sin 20°40' = 0.3529$)	**1**
$BD = 2 \times BC = 31.76$ m (since $DC = BC$)	**1**
(ii)	
In $\triangle DBE$ ($\angle B = 90°$): $\tan D\hat{E}B = \frac{BD}{BE}$	**1**
$= \frac{31.76}{45} = 0.7058$ ($= 2 \sin 20°40' = 0.7058$)	**1**
(iii)	
Tangent table: $\tan 35°10' = 0.7046$ and $\tan 35°20' = 0.7089$	**1**
$0.7058 - 0.7046 = 0.0012$; the difference per $10'$ is 0.0043, so the extra angle $\approx 3'$	**1**
$D\hat{E}B \approx 35°13'$ (accept $35°10'$ to $35°20'$)	**1**
(iv)	
$D\hat{E}B - C\hat{A}B = 35°13' - 20°40' = 14°33'$ (or: $D\hat{E}B > 35°10'$, so $D\hat{E}B - C\hat{A}B > 14°30'$)	**1**
$14°33' > 14°$, so the difference is greater than $14°$.	**1**

Total marks: 10

Question 4

(i)	
Area of the rectangle = $2x(x + 4) = 2x^2 + 8x$	**1**
Area of the hole = $(x - 2)^2 = x^2 - 4x + 4$	**1**
$2x^2 + 8x - (x^2 - 4x + 4) = 152$, so $x^2 + 12x - 156 = 0$	**1**
(ii)	
$x = \frac{-12 \pm \sqrt{(144 + 624)}}{2}$	**1**
$\sqrt{768} = \sqrt{(256 \times 3)} = 16\sqrt{3}$, so $x = -6 \pm 8\sqrt{3}$	**1**
x must be positive: $x = -6 + 8 \times 1.73 = 7.84$	**1**
$x \approx 8$ cm	**1**
(iii)	
$2x = 2(-6 + 8\sqrt{3}) = -12 + 16\sqrt{3}$	**1**
Using $\sqrt{3} = 1.73$: $16\sqrt{3} = 27.68$	**1**
$2x = 27.68 - 12 = 15.68$ cm > 15 cm (since $\sqrt{3} > 1.73$ the true value is even larger)	**1**

Total marks: 10

Question 5

(i)	
$x + y = 30$ ..... (1)	**1**
$90x + 140y = 3\,250$, i.e. $9x + 14y = 325$ ..... (2)	**1**
(1) × 9: $9x + 9y = 270$ ..... (3)	**1**
(2) − (3): $5y = 55$, so $y = 11$	**1**
$x = 30 - 11 = 19$. Small notebooks 19, large notebooks 11 (check: $90 \times 19 + 140 \times 11 = 1\,710 + 1\,540 = 3\,250$)	**1**
(ii)	
Let a small notebook have n pages; a large one has $n + 40$ pages	**1**
$19n + 11(n + 40) = 2\,840$	**2**
$30n = 2\,840 - 440 = 2\,400$	**1**
$n = 80$ pages	**1**

Total marks: 10

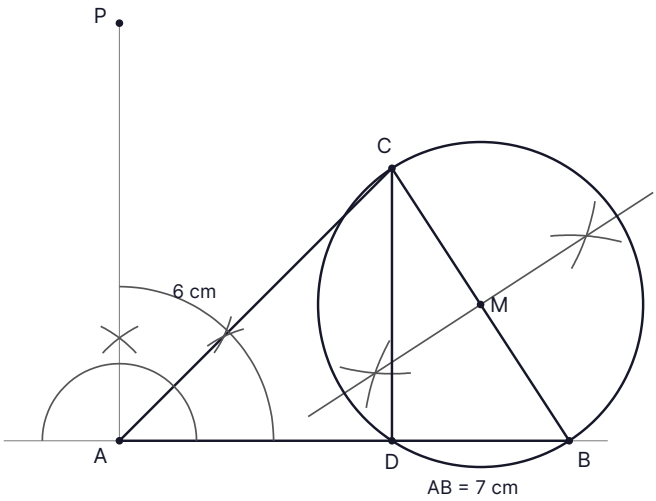
Question 6

(i)	
Mid-values: 24.5, 34.5, 44.5, 54.5, 64.5, 74.5 (in order)	**1**
$\Sigma fx = 2\times24.5 + 8\times34.5 + 10\times44.5 + 10\times54.5 + 8\times64.5 + 2\times74.5 = 49 + 276 + 445 + 545 + 516 + 149 = 1\,980$	**2**
Mean = 1 980 ÷ 40 = 49.5 customers	**1**
(ii)	
49.5 × 30	**1**
= 1 485 customers	**1**
(iii)	
49.5 × 7 × 120	**1**
= Rs 41 580	**1**
(iv)	
Days with fewer than 40 customers: 2 days in 20 – 29 and 8 days in 30 – 39	**1**
Minimum total = 2 × 20 + 8 × 30 = 40 + 240 = 280	**1**

Total marks: 10

Part B — Answers

Question 7

(i)	
 A correct construction (shown reduced in size)	
AB = 7 cm drawn (6.9 – 7.1 cm accepted)	**1**
90° at A constructed with arcs visible (perpendicular to AB at A)	**1**
(ii)	
Bisector of ∠BAP (45°) constructed with arcs	**1**
C marked with AC = 6 cm (5.9 – 6.1 accepted) and BC joined; BC ≈ 5.1 cm (5.0 – 5.2 accepted)	**1**
(iii)	
Arcs from B and C with equal radius (more than half of BC) drawn on both sides	**1**
Perpendicular bisector drawn; midpoint M of BC found	**1**
Circle with centre M and radius MB (≈ 2.5 cm) passing through B and C	**1**
(iv)	
D marked on AB (AD ≈ 4.2 cm) and CD joined	**1**
∠BDC = 90° (angle in a semicircle; BC is a diameter)	**1**
(v)	
In △ADC: ∠DAC = 45°, ∠ADC = 90° (angle sum 180°), so ∠ACD = 45°	**1**

Total marks: 10

Question 8

(i)	
In $\triangle GDE$ and $\triangle FCE$: $\angle GDE = \angle FCE$ (alternate angles, AG ∥ BF, DC the transversal)	**1**
$\angle DEG = \angle CEF$ (vertically opposite angles); DE = EC (E is the midpoint of DC)	**1**
$\therefore \triangle GDE \cong \triangle FCE$ (angle, angle, side)	**1**
(ii)	
From the congruence, GE = EF (corresponding sides)	**1**
In GDCF the diagonals GF and DC bisect each other (GE = EF and DE = EC), so GDCF is a parallelogram	**1**
(iii)	
In $\triangle HBF$ and $\triangle FCE$: $\angle HBF = \angle ECF$ (corresponding angles, HB ∥ EC)	**1**
$\angle HFB = \angle EFC$ (common angle); so the third angles are equal and $\triangle HBF$ is equiangular with $\triangle FCE$	**1**
$\triangle FCE \cong \triangle GDE$ (part (i)), so $\triangle HBF$ and $\triangle GDE$ are equiangular	**1**
(iv)	
$\triangle HBF \sim \triangle ECF$: $\frac{HB}{EC} = \frac{BF}{CF}$. $BF = BC - CF = \frac{2}{3}$ BC, so BF : CF = 2 : 1	**1**
$HB = 2 \times EC = 2 \times \frac{1}{2}$ DC = DC = AB (opposite sides of the parallelogram), so BH = AB	**1**

Total marks: 10

Question 9

(i)	
Volume of one pyramid = $\frac{1}{3} \times a^2 \times 3a = a^3$	**1**
Volume of 8 pyramids = $8a^3$; volume of the cylinder = $\pi r^2 \times 6r = 6\pi r^3$	**2**
$8a^3 = 6\pi r^3$, so $a^3 = \frac{3}{4}\pi r^3$	**1**
(ii)	
$a^3 = \frac{3 \times 3.14 \times 4.5^3}{4}$; $\lg a^3 = \lg 3 + \lg 3.14 + 3 \lg 4.5 - \lg 4$	**1**
$= 0.4771 + 0.4969 + 3 \times 0.6532 - 0.6021 = 0.4771 + 0.4969 + 1.9596 - 0.6021 = 2.3315$ (tables: lg 3 = 0.4771, lg 3.14 = 0.4969, lg 4.5 = 0.6532, lg 4 = 0.6021)	**1**
a^3 = antilog 2.3315 ≈ 214.5 (accept 214 – 215); lg a = 2.3315 ÷ 3 = 0.7772, a ≈ 5.99, so a ≈ 6 cm	**1**
(iii)	
Volume of the cuboid = $a^2 \times 3a = 3a^3$; volume of the pyramid = a^3	**1**
Remaining volume = $3a^3 - a^3 = 2a^3$	**1**
$= 2 \times 6^3 = 432$ cm^3 (accept 429 – 432 using $a^3 \approx 214.5$)	**1**

Total marks: 10

Question 10

(i)	
8, 12, 16	**1**
(ii)	
$T_n = a + (n - 1)d$: $T_{20} = 8 + 19 \times 4$	**1**
= 84 bottles	**1**
(iii)	
$S_{20} = \frac{20}{2}(8 + 84)$	**1**
= 920 bottles	**1**
(iv)	
$S_{16} = \frac{16}{2}[2a + 15 \times 7] = 920$	**1**
$2a + 105 = 115$	**1**
$a = 5$ bottles	**1**
(v)	
$8 + 4(n - 1) = 5 + 7(n - 1)$	**1**
$3 = 3(n - 1)$, $n = 2$: the 2nd day (both collected 12 bottles)	**1**

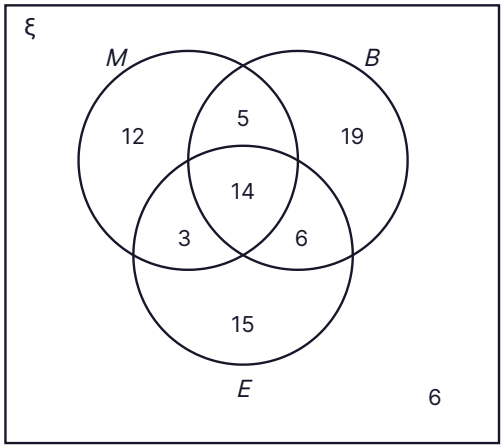
Total marks: 10

Question 11

(i)(a)	
$\angle CPB = 90°$ (CP ⊥ tangent PBQ) and $\angle CXB = 90°$ (given, CD ⊥ AB)	**1**
$\angle PBX = 90°$ (the tangent is perpendicular to the diameter AB at B)	**1**
In PBXC, $\angle CPB + \angle CXB = 90° + 90° = 180°$ (opposite angles supplementary), so PBXC is a cyclic quadrilateral	**1**
(i)(b)	
CP ⊥ PQ and AB ⊥ PQ, so CP ∥ AB; $\angle PCB = \angle CBX$ (alternate angles)	**1**
OB = OC (radii), so $\angle OCB = \angle OBC$ (base angles of an isosceles triangle), and $\angle OBC = \angle CBX$	**1**
$\therefore \angle PCB = \angle OCB$, so BC bisects $\angle PCO$	**1**
(ii)	
$\angle DBQ = \angle BAD$ (angle between the tangent BQ and chord BD equals the angle in the alternate segment)	**2**
$\angle DBQ = \angle BCD$ (angle between the tangent and chord BD equals the angle in the alternate segment, or $\angle BCD = \angle BAD$ angles in the same segment)	**2**

Total marks: 10

Question 12

(i)	
 The completed Venn diagram (shown reduced in size)	
Customers who bought at least one item = 80 − 6 = 74	**1**
At least two items: 28, so exactly one item = 74 − 28	**1**
= 46 customers	**1**
(ii)	
Let the number who bought all three = number who bought exactly two = k	**1**
k + k = 28	**1**
k = 14 customers bought all three	**1**
(iii)	
Exactly two items = 14: only (M, B) = 5, so only (M, E) + only (B, E) = 9. Let only (M, E) = e, then only (B, E) = 2e; 3e = 9	**1**
Only milk powder and eggs = 3; only bread and eggs = 6	**1**
(iv)	
Only milk powder = 34 − (5 + 3 + 14) = 12; only eggs = 38 − (3 + 6 + 14) = 15	**1**
Only bread = 46 − 12 − 15 = 19 customers	**1**

Total marks: 10